Satyam Patel

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EDUCATION

Stevens Institute of Technology | Hoboken, NJ | GPA: 3.2 / 4.0
Bachelor of Engineering in Mechanical Engineering| Minor in Systems and Industrial Engineering | Expected May 2029

Jefferson Township High School | Oak Ridge, NJ | GPA: 3.85 / 4.0
STEM Academy Graduate | National Honor Society | June 2025

TECHNICAL SKILLS

- **Design & Simulation:** SolidWorks 2025, Onshape, Canva
- **Manufacturing & Machining:** Precision Machining, Part Inspection, Quality Control, Manual Lathes, Milling, Welding, 3D Printing (Bambu Lab P2S, Maintenance & Troubleshooting), Cura Slicer
- **Software & Systems:** Arduino IDE, Google Workspace, Microsoft 365, Windows/Mac OS, Cloud Data Streaming
- **Soft Skills:** Project Management, Cross-functional Team Leadership, Technical Communication

EXPERIENCE & INTERNSHIPS

Precision Forms, Butler NJ - Intern | Precision Machining & Quality Control | May 2026 – August 2026

- Performed precision machining, thorough part inspection, and quality control on high-precision mechanical components using micrometers and monitoring strict tolerances as low as to five tenths of an inch.
- Responsible for the manufacturing processes for aerospace and medical grade components
- Collaborated on fabrication tasks and successfully executed a final capstone project involving the custom-machining of a metal barrel for a Pentel P207 mechanical pencil, focusing on tight-tolerance turning and assembly design.

ENGINEERING PROJECTS

Smart Autonomous Watering System | Electrical & Product Designer | Sept 2025 – Dec 2025

- Developed a self-contained IoT device to monitor soil moisture, light, and humidity using Arduino microcontrollers.
- Designed a custom modular housing in SolidWorks 2025 to integrate all mechanical and electronic components.
- Programmed logic for real-time data streaming to a cloud server while simultaneously managing pump actuation.
- Optimized system code for power efficiency and reliability during continuous 24/7 operation.

Vex Robotics Team | Lead Engineer & Captain | Sept 2023 – June 2025

- Head Lead of design and assembly of competitive robots, ensuring compliance with all Vex technical regulations.
- Mentored teammates on mechanical assembly and iterative design processes to solve field challenges.
- Coordinated with coaches to manage club fundraising and equipment procurement.

Drone Racing Club | Vice President | Sept 2020 – June 2025

- Managed equipment inventory and performed technical troubleshooting/repairs on racing drones.
- Facilitated training sessions for new members on flight mechanics and safety protocols.
- Collaborated with the President to plan meeting agendas and coordinate club events.

LEADERSHIP

Boy Scouts of America (Patriots Path Council) | Eagle Scout | June 2018 – Present

- Achieved Eagle Scout rank (attained by only 6% of Scouts), completing a comprehensive community service capstone.
- Order of the Arrow: Inducted into BSA's National Honor Society for exemplary service and leadership.

AWARDS & CERTIFICATIONS

- Eagle Scout Rank (2025)
- National Youth Leadership Training (NYLT) Graduate
- Sea Base Captains Club Award (Recognized for leadership and teamwork in high-adventure marine environments)